MTH 201 Chapter 2: Integration Techniques

2.1 Integration by Parts

General Idea:
We want to use the product rule to develop a useful formula for integration.

Formula (Integration by parts):

Example 1:

(a) Find $\int x\cos(x)\,dx$.

(b) If you let $u=\cos(x)$ & $dv=x\,dx$ what does the integration by parts formula yield?

Example 2:

Find $\int \ln(x)\,dx$.

Note: Sometimes we need to apply the integration by parts formula more than one time.

Example 3:

Find $\int x^2 \sin(x)\,dx$.

Theorem (Tabular Integration by Parts):

Visualizing Tabular Integration by Parts:

Consider the integral: $\int x^2 \sin(x)\,dx$.

Example 4 (you try):

Find $\int (x^2+1)e^{-x}\,dx$.

Formula (Integration by Parts for Definite Integrals): If f' and g' are continuous on $[a,b]$, then

$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big|_a^b - \int_a^b g(x)f'(x)\,dx$$

Example 5:

Calculate the exact value of: $\int_1^2 [\ln(x)]^2\,dx$

Case of the Unknown Integral

Example 6:

Find $\int e^x \cos(x)\,dx$.